MDEF: Multi-Report Deep Ensemble Fusion for 3D CT Report Generation

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Code and paper

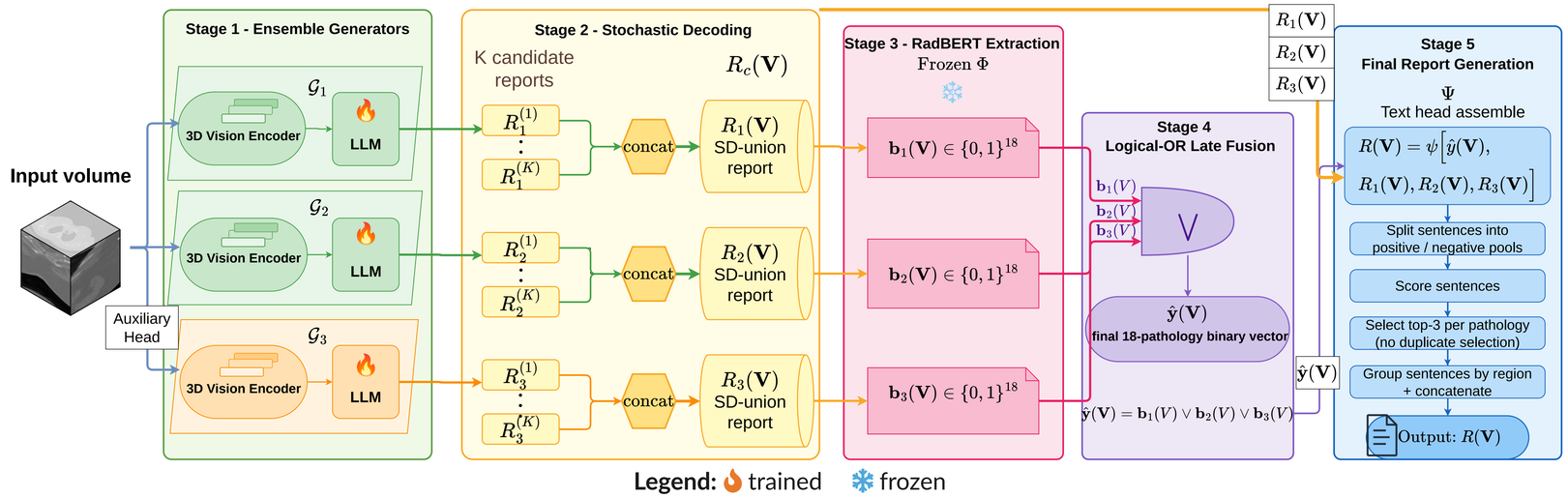
1 Background and Challenges

- **The single-report paradigm misses pathologies:** existing 3D CT systems decode one report per volume by greedy or beam search, so many of the 18 pathologies are never mentioned. Under clinical-efficacy (CE) F1 this shows as high precision with low recall. On RG, Reg2RG reaches precision **0.44** and recall **0.18**.
- **Generalization makes it harder:** the same recall limitation grows once evaluation moves to different datasets.

2 Contributions

- **A deep-ensemble architecture.** We propose MDEF, a five-stage pipeline that aggregates multiple generated reports at the 18-pathology binary-vector level by logical-OR, so any pathology mentioned in any sampled report reaches the final report. To the best of our knowledge, MDEF is the first such pipeline for 3D CT report generation.
- **A generalization evaluation.** We train MDEF on RadGenome and evaluate it under one fixed configuration on three different held-out datasets. Our MDEF reaches the highest CE F1 on all three.

3 Overall Architecture



- **Stage 1 and Stage 2:** three generators each draw $K{=}10$ Stochastic Decoding candidates that are concatenated into one SD-union report.
- **Stage 3:** RadBERT maps every SD-union report to an 18-pathology binary vector.
- **Stage 4:** a per-pathology logical-OR fuses the three vectors, so the ensemble misses a pathology only when every generator misses it.
- **Stage 5, Text head assemble:** the deterministic text head scores candidate sentences, selects the highest scoring ones per pathology and groups them by anatomical region.

4 Experimental Setup

- **Trained on RadGenome (RG) only,** with one fixed configuration applied unchanged to every held-out dataset.
- **Three held-out datasets:** RadGenome (RG), the training distribution. CT-RATE (CTR), scored against its official human labels. INSPECT (INS), from a different institution.
- **Metrics:** clinical-efficacy (CE) precision, recall and F1 over the 18 CT-RATE chest abnormality labels, read by RadBERT on RG and INS. Plus BLEU, METEOR and ROUGE-L.
- **Baselines:** Reg2RG, CT-GRAPH and MARCH, the strongest single-report generators.
- **Configuration:** our MDEF and all three baselines share a 3D vision encoder with Llama-2-7B as the language model.

5 Empirical Results

	RG			CTR			INS		
CE	P	R	**F1**	P	R	**F1**	P	R	**F1**
Reg2RG	**.442**	.181	.256	**.322**	.069	.113	.128	.138	.133
CT-GRAPH	.425	.177	.250	.313	.062	.103	.127	.124	.126
MARCH	.328	.316	.321	.300	.287	.293	**.131**	.259	.174
Our MDEF	.319	**.537**	**.400**	.255	**.501**	**.338**	.110	**.510**	**.181**

Bold is the best value in each column.

- **Highest CE F1 on all three datasets,** beating MARCH by **+0.079**, **+0.045** and **+0.008**.
- **Recall and precision:** recall rises by **+70%** to **+97%** over MARCH, and precision stays within **0.05** of MARCH on all three datasets.
- **Per-pathology recall:** higher than MARCH on **18 of 18** pathologies on RG, **14 of 18** on CTR and **13 of 18** on INS.
- **NLG metrics:** best on **9 of the 12** BLEU, METEOR and ROUGE-L scores on CTR and INS.
- **Inference cost:** **13%** lower wall time than MARCH, at the same 7.0B $\times$ 3 parameter budget.

6 Qualitative Evidence



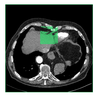
RG / pericardial effusion

GT. "Pericardial effusion at the apex, ~12 mm. No pneumothorax."
Reg2RG. "Mediastinal contours unremarkable."
CT-GRAPH. "Heart size within normal limits. Lungs clear."
MARCH. "No pneumothorax. Mediastinum and hila unremarkable."
Our MDEF. "12 mm pericardial effusion observed at the apex."



CTR / atelectasis

GT. "Atelectatic changes in the lower lobes. No pleural effusion."
Reg2RG. "Mediastinum and hila unremarkable."
CT-GRAPH. "Lung parenchyma is otherwise clear."
MARCH. "No pleural effusion. Heart size within normal limits."
Our MDEF. "Atelectasis observed in the lower-lobe posterobasal segment."



INS / hiatal hernia

GT. "Large hiatal hernia, intrathoracic stomach. No pulmonary embolus."
Reg2RG. "No pulmonary embolus. Heart size normal."
CT-GRAPH. "Lungs clear, no acute thoracic pathology."
MARCH. "No pleural effusion. No pneumothorax."
Our MDEF. "Hiatal hernia is observed."

7 Conclusion

- **Method:** MDEF turns the single-report paradigm into a multi-report one and fuses reports at the 18-pathology vector level.
- **Performance:** our MDEF reaches the highest CE F1 on all three held-out datasets under one fixed configuration.
- **Recall gain:** the improvement is driven by a per-pathology recall gain over every single-report baseline.
- **Future work:** addressing the precision cost of the logical-OR aggregator, and broadening the evaluation to more chest-CT datasets.

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